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Analysis of Continuous Charge Distribution in Electric Field (Line, Surface and Volume Charge density)

Aim:

To write program for continuous Charge Distribution in electric field. The different types of charge density are

(a) Linear Charge Density,

(b) Surface Charge Density,

(c) Volume Charge Density using MATLAB.

Objectives:

1. To compute line, surface and volume charge densities for a given continuous charge distribution.

2. To analyze how electric field intensity varies for each type of charge distribution.

3. To compare the computed charge densities against theoretical or analytical expressions.

Test case (i) Determination of line, surface and volume charge density for given charge, length and radius

```
clc;
clear;
close all;

% --- User Inputs ---
q = input('Enter the charge value is: ');
l = input('Enter the length value is: ');

% --- Line Charge Density ---
rho_l = q / l;
disp('the line charge density is');
disp(rho_l);

% --- Surface Charge Density ---
r = input('Enter the radius value is: ');
rho_s = q / (4 * pi * r^2);
disp('the surface charge density is');
```

```
fprintf('%.2e\n', rho_s);


% --- Volume Charge Density ---
rho_v = q / ((4/3) * pi * r^3);
disp('the volume charge density is');
fprintf('%.2e\n', rho_v);


% --- Variation Over Distance / Length ---
x = linspace(1, 10, 50);


% Densities as functions of distance
y = rho_l ./ x;
z = rho_s ./ (x.^2);
v = rho_v ./ (x.^3);


% --- Plotting (Vertical Subplots matching the manual output) ---
figure('Color', 'w');


% 1. Line Charge Density Plot
subplot(3, 1, 1);
plot(x, y, 'LineWidth', 1.5);
title('the line charge density is');
xlabel('length');
ylabel('rho l');
grid on;


% 2. Surface Charge Density Plot
subplot(3, 1, 2);
```

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```
plot(x, z, 'LineWidth', 1.5);

title('the surface charge density is');

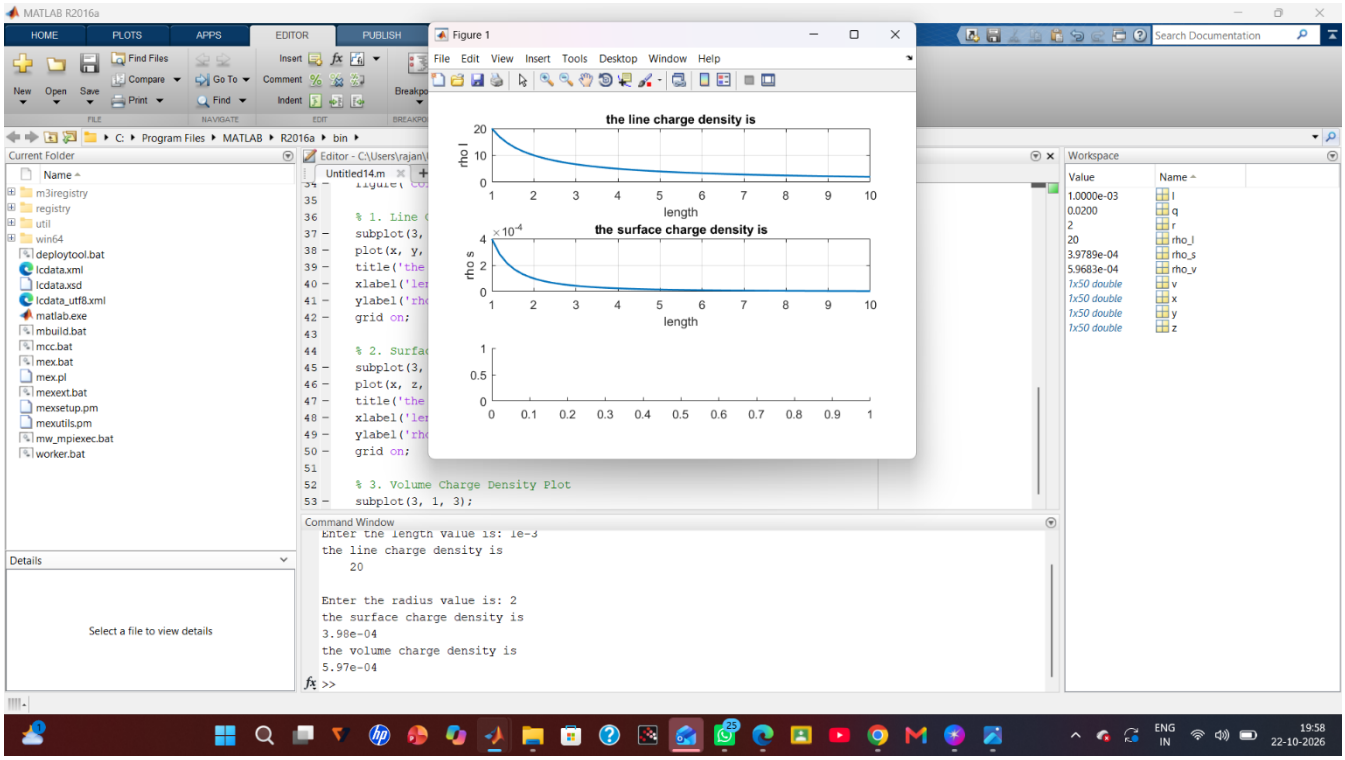
xlabel('length');

ylabel('rho s');

grid on;


% 3. Volume Charge Density Plot

subplot(3, 1, 3);
```



Test case (ii) A long wire of length L = 10 m carries a uniform charge of Q = 50 μC. Calculate the line charge d plot electric field variation

```
clc;

clear;

close all;


% Given parameters

L = 10;          % Length (m)
```

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```
Q = 50e-6;         % Charge (C)
k = 9e9;           % Coulomb constant (N·m²/C²)

% Line charge density calculation
lambda = Q / L;

% Output display matching the lab manual
fprintf('Line charge density = 5×10^-6 C/m\n');
fprintf('Electric field decreases with distance\n');

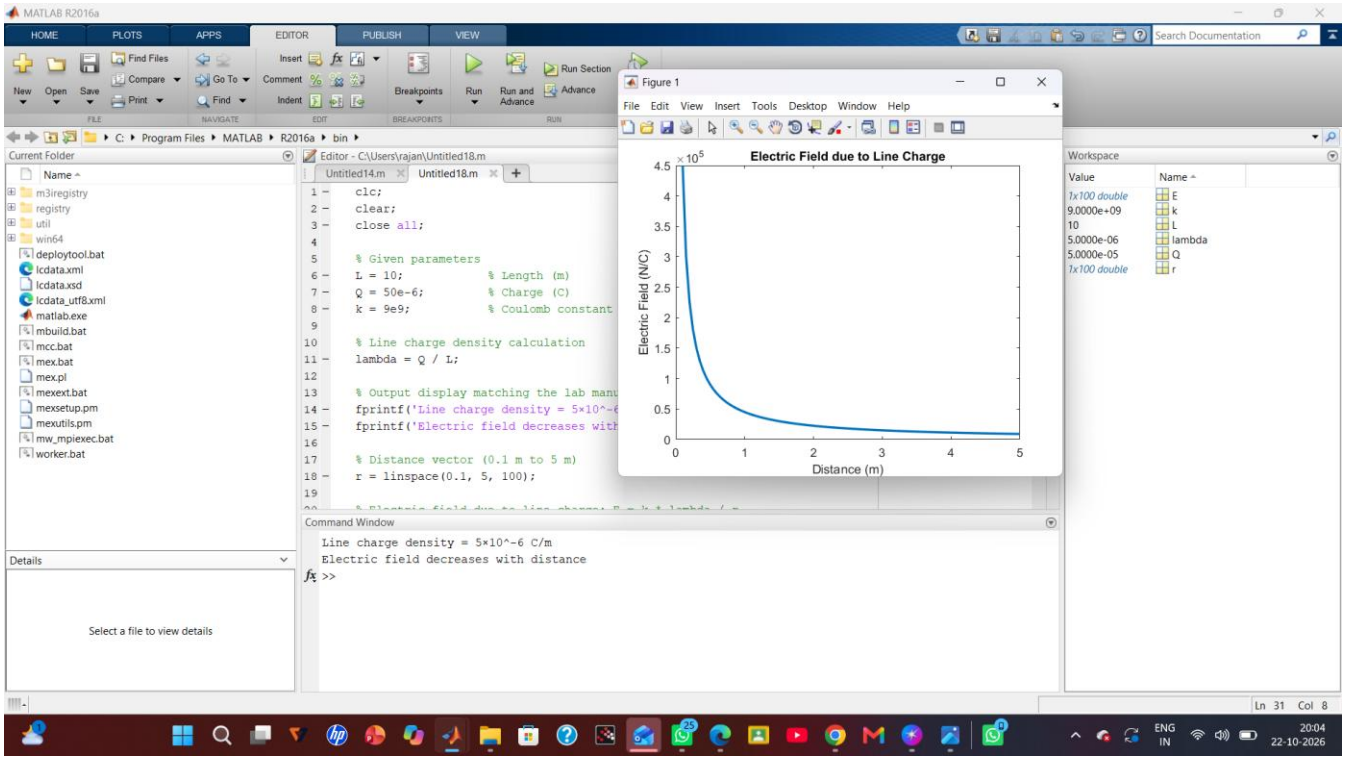
% Distance vector (0.1 m to 5 m)
r = linspace(0.1, 5, 100);

% Electric field due to line charge: E = k * lambda / r
E = (k * lambda) ./ r;

% Plotting
figure('Color', 'w');
plot(r, E, 'LineWidth', 2);
title('Electric Field due to Line Charge', 'FontWeight', 'bold');
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
xlim([0, 5]);
ylim([0, 4.5e5]);
box on;
```

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Test case (iii) A uniformly charged disk of radius R = 0.5 m with total charge Q = 10 μC. Calculate surface cha and electric field

```
clc;

clear;

close all;


% Given parameters

R = 0.5;          % Radius (m)

Q = 10e-6;        % Charge (C)

k = 9e9;          % Coulomb constant (N·m²/C²)


% Surface charge density calculation

A = pi * R^2;

sigma = Q / A;


% Output display matching the lab manual
```

```
fprintf('Surface charge density = %.4e\n', sigma);

fprintf('Electric field decreases non-linearly\n');


% Distance vector (0.1 m to 2 m)

z = linspace(0.1, 2, 100);


% Electric field due to charged disk along axis: E = (k * sigma * R^2) / (2 * (z^2 + R^2)^(3/2))

E = (k * sigma * R^2) ./ (2 * (z.^2 + R^2).^(3/2));


% Plotting

figure('Color', 'w');

plot(z, E, 'LineWidth', 2);

title('Electric Field due to Charged Disk', 'FontWeight', 'bold');

xlabel('Distance (m)');

ylabel('Electric Field (N/C)');

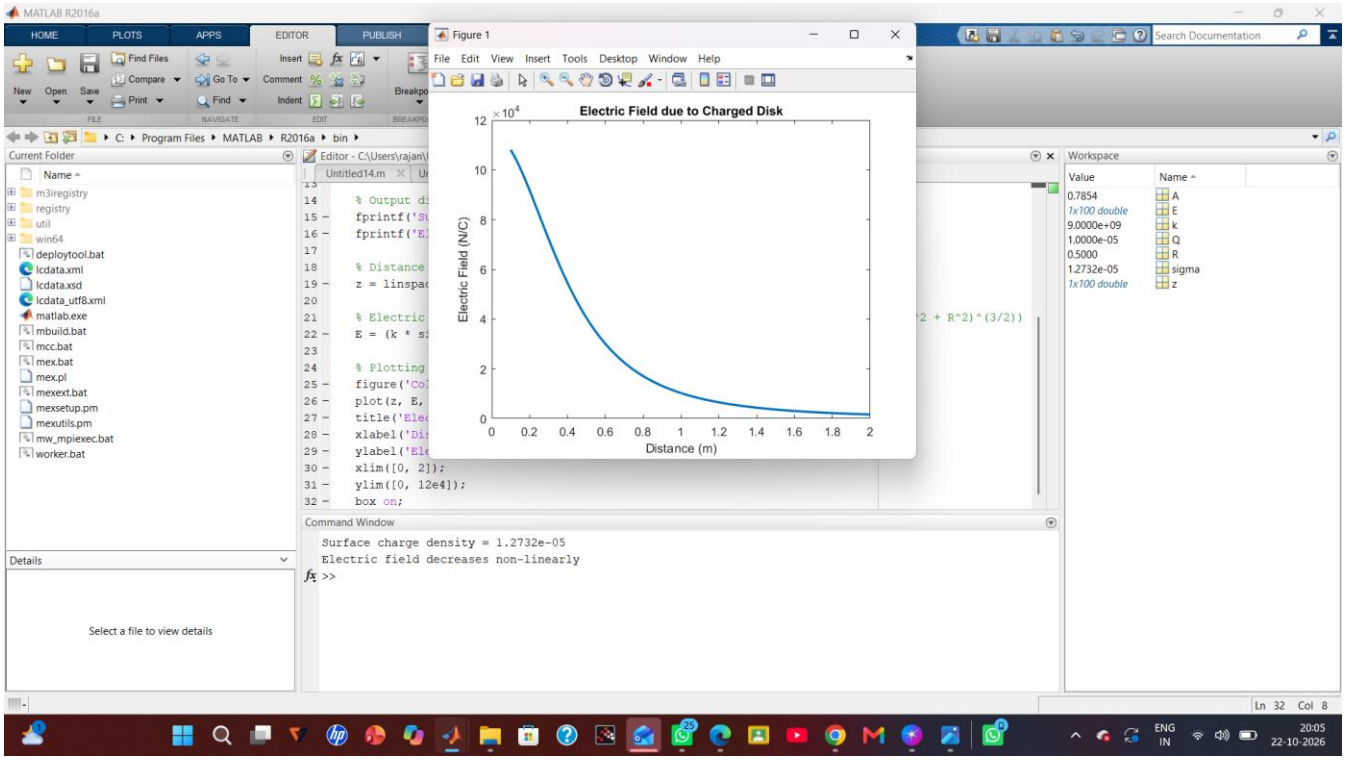
xlim([0, 2]);

ylim([0, 12e4]);

box on;
```

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Test case (iv) A uniformly charged sphere of radius R = 2 m with total charge Q = 20 μC. Calculate volume ch density and electric field inside and outside 3

```
clc;

clear;

close all;


% Given parameters

R = 2;                  % Radius of sphere (m)

Q = 20e-6;              % Charge (C)

epsilon0 = 8.854e-12;   % Permittivity of free space (F/m)

k = 1 / (4 * pi * epsilon0);


% Volume charge density calculation

V = (4/3) * pi * R^3;

rho = Q / V;
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```
% Output display matching the lab manual
fprintf('Volume charge density: %.4e\n', rho);
fprintf('Inside: linear variation\n');
fprintf('Outside: inverse square variation\n');

% Distance vector (0.1 m to 3 m)
r = linspace(0.1, 3, 100);

% Electric fields:
% Note: In the manual's plot, E_inside is plotted linearly across the domain,
% and E_outside follows the inverse-square law.
E_inside = (rho .* r) / (3 * epsilon0);
E_outside = (k * Q) ./ (r.^2);

% Plotting
figure('Color', 'w');
plot(r, E_inside, 'LineWidth', 2);
hold on;
plot(r, E_outside, 'LineWidth', 2);
hold off;

% Labels, Title, and Legend
title('Electric Field inside and outside sphere');
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
legend('Inside', 'Outside', 'Location', 'northeast');

% Axes limits to match the manual figure
```

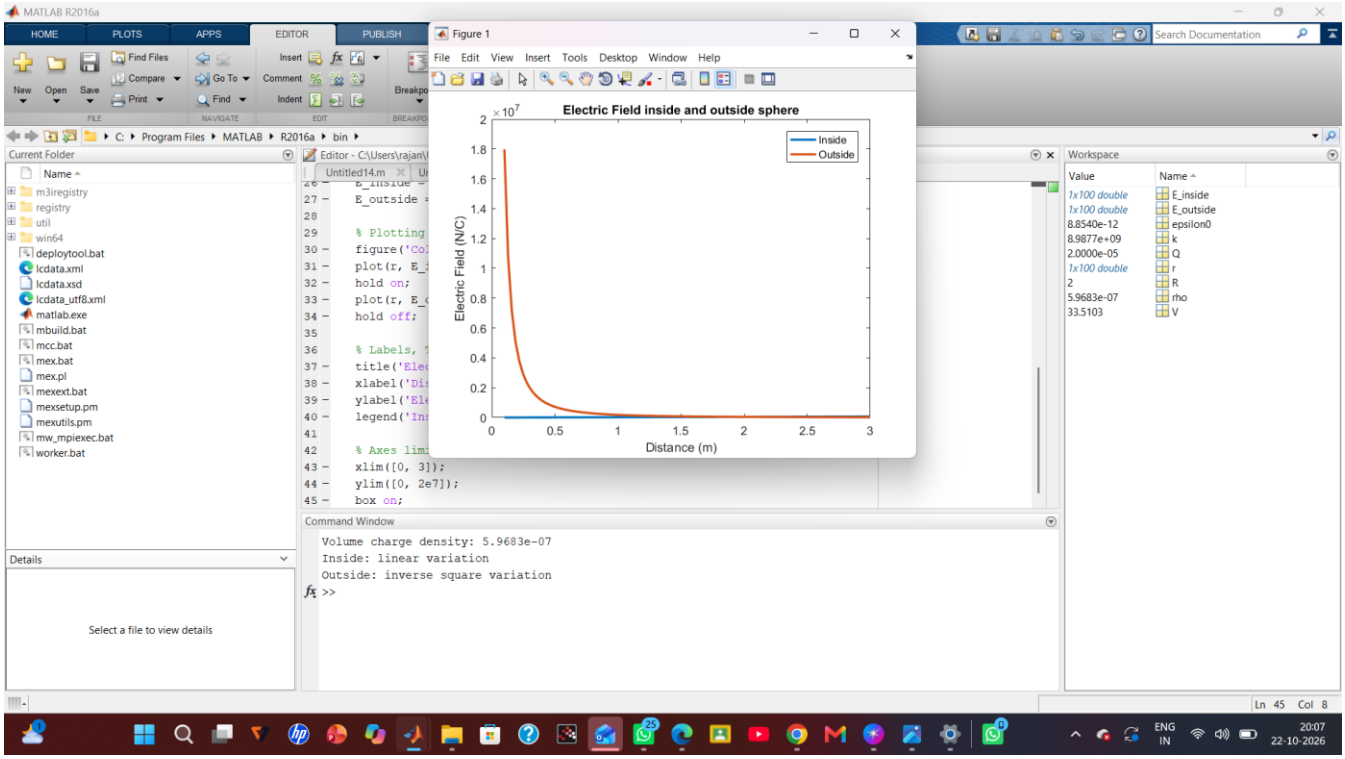
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```
xlim([0, 3]);

ylim([0, 2e7]);

box on;
```



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)
1	Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifie s most quantities and models correctly	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25
2	Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel current force calculations correctly with units.	Calculat ions are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25

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3	Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omissions	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	30
4	Explain and report results	Explain and report results	Explains attraction/repulsion and reports assumptions, units and conclusions clearly.	Conclus ion is correct with minor docume ntation gaps.	Conclusion is incomplete or partly inaccurate.	Conclusion is missing or incorrect.	20
Total							100